

Experiment 14

14.1.1 Complex Number Addition using Class

Algorithm :

Step 1: Start

Step 2: Define class Complex with three methods:

- initComplex(self)
- sum(self, c1, c2)
- display(self)

Step 3: Method initComplex(self)

- 3.1 Read two integers from input
- 3.2 Assign first value to self.real
- 3.3 Assign second value to

self.img Step 4: Method sum(self, c1, c2)

- 4.1 Set self.real = c1.real + c2.real
- 4.2 Set self.img = c1.img +

c2.img Step 5: Method display(self)

- 5.1 If self.img >= 0:
Print self.real + self.img i
- 5.2 Else:
Print self.real -

|self.img| i Step 6: Create three objects

- 6.1 c1 = Complex()
- 6.2 c2 = Complex()
- 6.3 c3 = Complex()

Step 7: Initialize c1 and c2

- 7.1 Call c1.initComplex()
Read and assign real and img for c1
- 7.2 Call c2.initComplex()
Read and assign real and img

for c2 Step 8: Compute sum

- 8.1 Call c3.sum(c1, c2)
- 8.2 c3.real = c1.real + c2.real
- 8.3 c3.img = c1.img + c2.img

Step 6: Display result

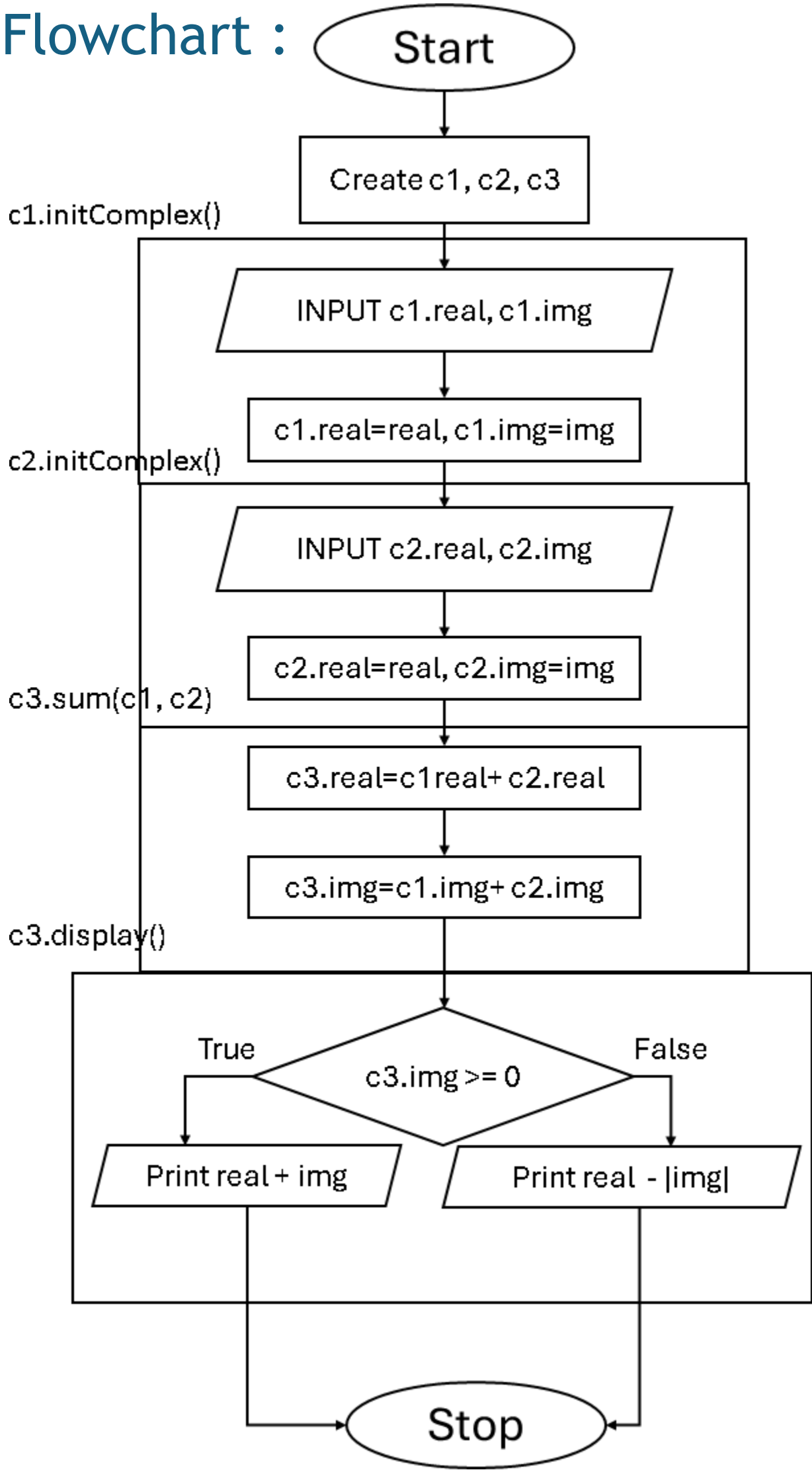
- 6.1 Call c3.display()
- 6.2 If c3.img >= 0:
Print c3.real + c3.img i

6.3Else:

Print c3.real - |c3.img|

i Step 10: Stop

Flowchart :



Code :

```

class Complex:
def
initComplex(self):

self.real , self.img = map(int,input().split())

def sum(self,c1,c2):
self.real = c1.real +
c2.real self.img = c1.img +
c2.img

def display(self):
if self.img >= 0:
print(f"{self.real} +
{self.img}i") else:
print(f"{self.real} - {abs(self.img)}i")
# Create three instances
c1          =
Complex()  c2
=  Complex()
c3          =
Complex()

# Initialize two complex
numbers c1.initComplex()
c2.initComplex()

# Compute and display sum
c3.sum(c1, c2)
c3.display()

```

Execution :

The screenshot displays the CodeTantra IDE interface. On the left, a sidebar lists course modules under 'Programming and Problem Solving Lab - TE7287 - II Sem - 2026'. The main editor area shows a Python program for adding two complex numbers using a class named `Complex`. The code includes methods for initialization, summation, and display. The instructions specify creating three instances of the `Complex` class, initializing two, and then computing and displaying the sum. The right-hand pane shows the code explorer with line numbers, the output area displaying the sum `5 + 9i`, and test results indicating that 4 out of 4 shown and hidden test cases passed.